Department of Computer Science

Faculty of Computing

# Project 2

| | | |
|---|---|---|
| **Programme** | : | Bachelor of Computer Science (*Data Engineering*) |
| **Subject Code** | : | SECP2753 |
| **Subject Name** | : | Data Mining |
| **Session-Sem** | : | 2024/2025-2 |


**Prepared by** :

1. LEE YIN SHEN (A23CS0236)
2. ELIJAH SHE YU SHENG (A23CS0073)
3. NEO LI XIN (A23CS0253)


**Section** : 02

**Lecturer** : Dr Rozilawati binti Dollah @ Md Zain

**Date** : 20-06-2025

# TABLE OF CONTENT

# 1.0 Introduction

For this assignment, we are tasked to collect and analyse the dataset chose to identify the potential business insights. We collected a dataset that represents the customer's shopping behaviour in the fashion industry. It provides comprehensive information about customer's product preferences and spending habits during their shopping experience. This dataset encompasses a range of variables, from customers profiling and clothing categories to payment details. We hope that through this dataset, we can clearly showcase the customers' shopping preferences with data visualization, thus aiding the company in identifying the market trends to increase the business sales and profits.

## 1.1 Overview of the dataset

The dataset mainly focuses on the customers' spending behaviour in the fashion industry. The dataset is sourced from Kaggle, a data science platform which contains various types of datasets. It was first published by Shahzad Aslam and the dataset is scheduled to be updated annually. This dataset contains 3900 rows and 21 columns. The dataset is fully completed, and no null or duplicate data are detected. Below is the list of data that are included in this dataset:

| No. | Feature Name | No. | Feature Name |
|---|---|---|---|
| 1 | Customer ID | 12 | Subscription Status |
| 2 | Age | 13 | Shipping Type |
| 3 | Gender | 14 | Discount Applied |
| 4 | Item Purchased | 15 | Promo Code Used |
| 5 | Category | 16 | Previous Purchases |
| 6 | Purchase Amount (USD) | 17 | Payment Method |
| 7 | Location | 18 | Frequency of Purchases |
| 8 | Size | 19 | Discount Percentage |
| 9 | Colour | 20 | Customer Income Level |
| 10 | Season | 21 | Item Price |
| 11 | Review Rating | | |

Table 1.1.1: Data Set List

## 1.2 Objective of the analysis

The primary goal of this analysis is to employ clustering models to understand and predict customer behaviour patterns. Specifically, the analysis aims to:

1. Classify customers based on age group dominance
   Identifying which age groups are more prevalent in the customer base and understanding their characteristics.
2. Determine payment method preferences
   Classifying customers based on their chosen method of payment and identifying demographic trends.
3. Assess the influence of income levels on purchasing behaviour
   Investigating whether income brackets significantly influence how often or how much customers spend.

By building and evaluating multiple clustering models, the analysis will reveal underlying trends and relationships that can inform marketing strategies, product placement, and customer experience optimization.

## 1.3 Business insights to be explored

Understanding customer behaviour is critical to developing data-driven business strategies. The analysis will aim to extract the following actionable insights:

1. Dominant Customer Demographics
   Identifying which age groups form the core of the customer base can help tailor products and marketing strategies more effectively.

2. Preferred Payment Channels
   Understanding payment preferences can guide financial partnerships (e.g., with digital wallet providers), streamline checkout processes, and increase customer satisfaction.

3. Income vs. Spending Behaviour
   Analysing the relationship between income and purchasing frequency or volume will help in segmenting customers by value and targeting high-value segments with loyalty programs or premium offerings.

These insights are crucial for businesses seeking to improve customer acquisition, retention, and overall satisfaction by aligning operational strategies with customer preferences and behaviours.

## 1.4 Clustering methods selected for the analysis

To carry out the clustering analysis, we selected two machine learning models: K-Means and DBSCAN. These algorithms were chosen for their ability to uncover patterns in the data and provide meaningful segmentation based on both numerical and categorical attributes.

### 1.4.1 K-Means Clustering

K-Means clustering is a widely-used partitioning method that divides data into a predefined number of clusters based on their proximity to the cluster centroids. It is known for its simplicity, speed, and effectiveness in handling large datasets. One of the main advantages of K-Means is its efficiency in grouping similar data points when the clusters are spherical and roughly equal in size. However, the limitation of K-Means is that it requires users to specify the number of clusters beforehand, which may not always be intuitive. Furthermore, it is sensitive to outliers and may not perform well when the clusters have irregular shapes or vary in density.

### 1.4.2 DBSCAN

DBSCAN, or Density-Based Spatial Clustering of Applications with Noise, is a density-based clustering algorithm that groups data points based on the density of their neighbourhoods. It does not require the number of clusters to be specified in advance and is capable of detecting clusters of arbitrary shape. This makes DBSCAN especially useful in identifying outliers and noisy data. Among its advantages are its flexibility and its effectiveness in dealing with datasets that contain noise or clusters of varying shapes. However, DBSCAN's performance relies heavily on the correct selection of parameters such as the neighbourhood's radius (Eps) and the minimum number of points (MinPts), and it may struggle with datasets containing clusters of varying densities.

# 2.0 Data Preparation

Before moving to the machine learning analysis, we do the data prerpocessing to ensure accurate results. The data preparation was done using Python in Google Colab, where we applied techniques like normalization, encoding, and selecting important features. These steps is to prepare a dataset that ready for analysis.

## 2.1 Data Normalization

```
scaler = MinMaxScaler()
dataset[['Purchase Amount (USD)', 'Item Price']] = scaler.fit_transform(dataset[['Purchase Amount (USD)', 'Item Price']])
```

We normalize the Purchase Amount (USD) and Item Price columns using Min-Max scaling to bring them within the same range of [0, 1]. This ensures that no feature dominates the clustering process due to its scale, allowing the classification algorithms to treat all features equally and improve the accuracy of clustering.

## 2.2 Data Encoding

```
# Use LabelEncoder to convert categorical columns into numeric format
label_encoder = LabelEncoder()

# Encoding Gender, Category, Subscription Status, Location, Payment Method, etc.
dataset['Gender'] = label_encoder.fit_transform(dataset['Gender'])
dataset['Category'] = label_encoder.fit_transform(dataset['Category'])
dataset['Subscription Status'] = label_encoder.fit_transform(dataset['Subscription Status'])
dataset['Location'] = label_encoder.fit_transform(dataset['Location'])
dataset['Payment Method'] = label_encoder.fit_transform(dataset['Payment Method'])
dataset['Frequency of Purchases'] = label_encoder.fit_transform(dataset['Frequency of Purchases'])
dataset['Customer Income Level'] = label_encoder.fit_transform(dataset['Customer Income Level'])
```

We encode the categorical columns like Gender, Category, Subscription Status, Location, Payment Method, Frequency of Purchases and Customer Income Level using Label Encoder. This converts each unique category into a numerical label, allowing these features to be used in the clustering model.

## 2.3 Data Export

```
output_file = 'preprocessed_data_full.csv'
dataset.to_csv(output_file, index=False)
```

We export the finalized dataset into a new CSV file. This output file contains all selected and transformed attributes in a structured format, including headers and comma-separated values, making it suitable for subsequent model training and evaluation.

# 3.0 Clustering Models

This section assesses two clustering algorithms, K-Means and DBSCAN, to derive actionable insights from consumer data in the fashion industry. The performance of these clustering models is evaluated by analysing intra-cluster cohesion, inter-cluster separation, and cluster quality.

## 3.1 Age Group Dominance

Age is a key demographic variable that significantly influences shopping behaviour, product preferences, and spending patterns. The "Age" column in the dataset provides valuable insights into customer segmentation based on generational tendencies. By applying clustering models to this feature, we can uncover which age groups are more actively engaged in online fashion retail, identify shared traits among them, and support data-driven decisions for age-specific marketing and product recommendations.

### 3.1.1 K-Means Clustering

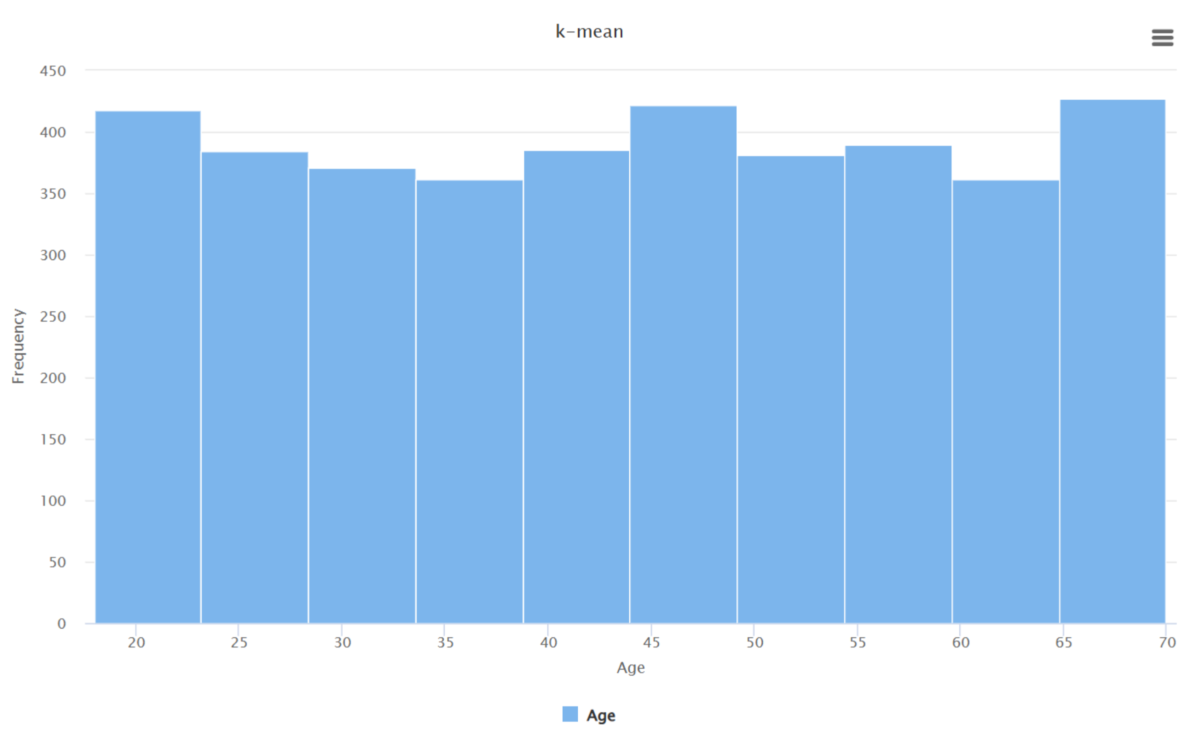


Figure 3.1.1.1 Histogram trained using K-Means Clustering

## PerformanceVector

```
PerformanceVector:
Avg. within centroid distance: 25.882
Avg. within centroid distance_cluster_0: 27.107
Avg. within centroid distance_cluster_1: 27.026
Avg. within centroid distance_cluster_2: 23.346
Davies Bouldin: 0.504
```

Using K-Means clustering, the dataset's "Age" column was segmented into three distinct clusters, revealing generational groupings such as younger adults which is Cluster 0, middle-aged consumers which is Cluster 1, and older adults which is Cluster 2. The clustering results show an average within-centroid distance of 25.882, with Cluster 2 being the most compact which is 23.346, and a low Davies-Bouldin index of 0.504, indicating well-separated and cohesive clusters. The histogram further supports a fairly uniform age distribution across the 18 to 70 range, with slight peaks at both ends. These insights support data-driven marketing strategies, enabling businesses to personalize campaigns, product recommendations, and engagement tactics for each age-based segment effectively.

### 3.1.2 DBSCAN

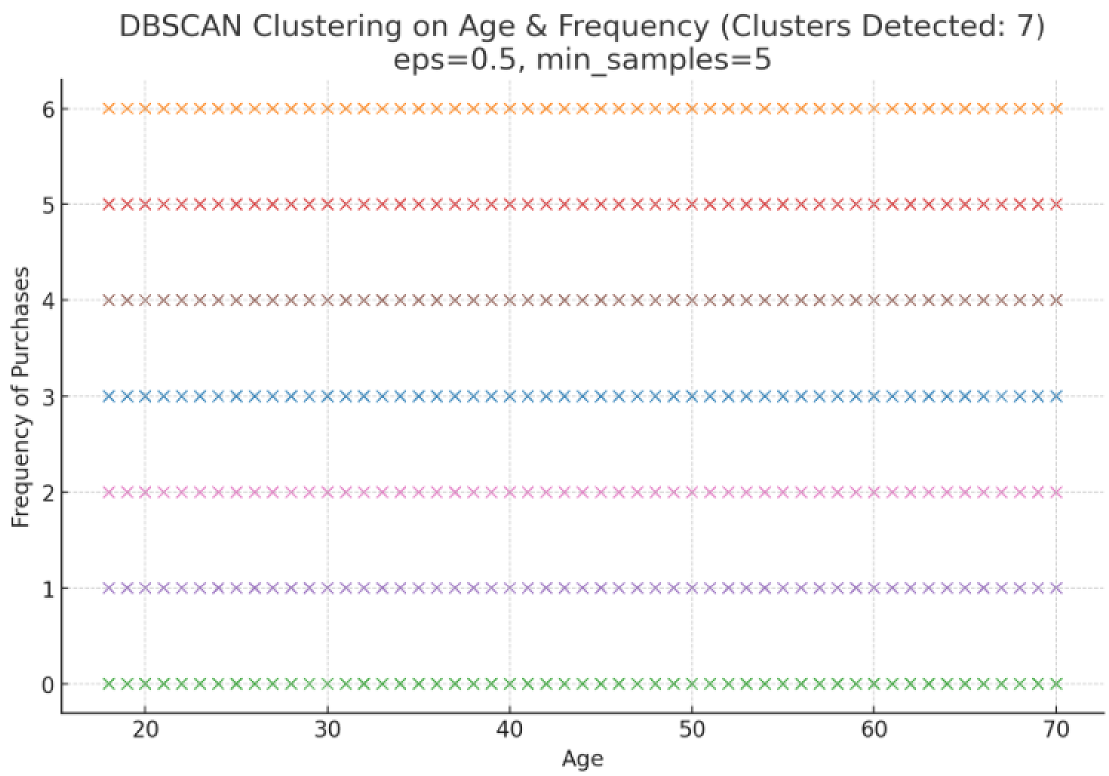


Figure 3.1.1.1 Scatterplot trained using DBSCAN Clustering

The scatter plot visualizes the results of DBSCAN clustering using two features: Age (x-axis) and Frequency of Purchases (y-axis). Each dot represents a customer, and colours indicate their assigned cluster. Notably, the points are aligned in horizontal rows, meaning customers with the same purchase frequency form distinct bands. The clustering results in 7 coloured bands, each corresponding to one of the 7 unique frequency values (0 through 6). The vertical spread (Age) within each band shows no clear separation by age, implying that DBSCAN formed clusters almost entirely based on frequency alone, not a combination of age and frequency.

```
Number of customers in each cluster (including noise):
Cluster
0    542
1    539
2    572
3    563
4    547
5    553
6    584
Name: count, dtype: int64

Average Age and Frequency per cluster (excluding noise):
           Age  Frequency of Purchases
Cluster
0        43.59                     3.0
1        44.65                     6.0
2        44.67                     0.0
3        44.69                     5.0
4        43.20                     1.0
5        44.28                     4.0
6        43.41                     2.0

Number of noise points (label = -1):
0
```

Figure 3.1.1.1 RESULT trained using DBSCAN Clustering

The clustering output identified 7 distinct clusters, each corresponding to a unique purchase frequency level, with average ages ranging narrowly between 43 and 45 years. This indicates that while frequency of purchases varied across customers, age did not significantly influence clustering. Each cluster contains a balanced number of customers (around 540 and 580), and there were no noise points (outliers). Although the clustering is technically correct, it reveals that DBSCAN did not find complex patterns involving both age and frequency rather, it simply grouped users with identical frequency values together, which limits its usefulness for deeper customer segmentation.

## 3.2 Payment Method Preferences

Payment method is important transactional component that affects how people buy. The dataset's "Payment Method" column provides a useful perspective on how various payment methods impact consumer buying habits. We can see clear patterns in the relationship between chosen payment methods and item selection, order value, and purchase frequency by studying data.

### 3.2.1 K-Means Clustering

```
Cluster distribution by Payment Method:
Cluster           0    1    2
Payment Method
Bank Transfer   196  218  198
Cash            211  227  232
Credit Card     218  234  219
Debit Card      216  219  201
PayPal          223  231  223
Venmo           219  220  195
```

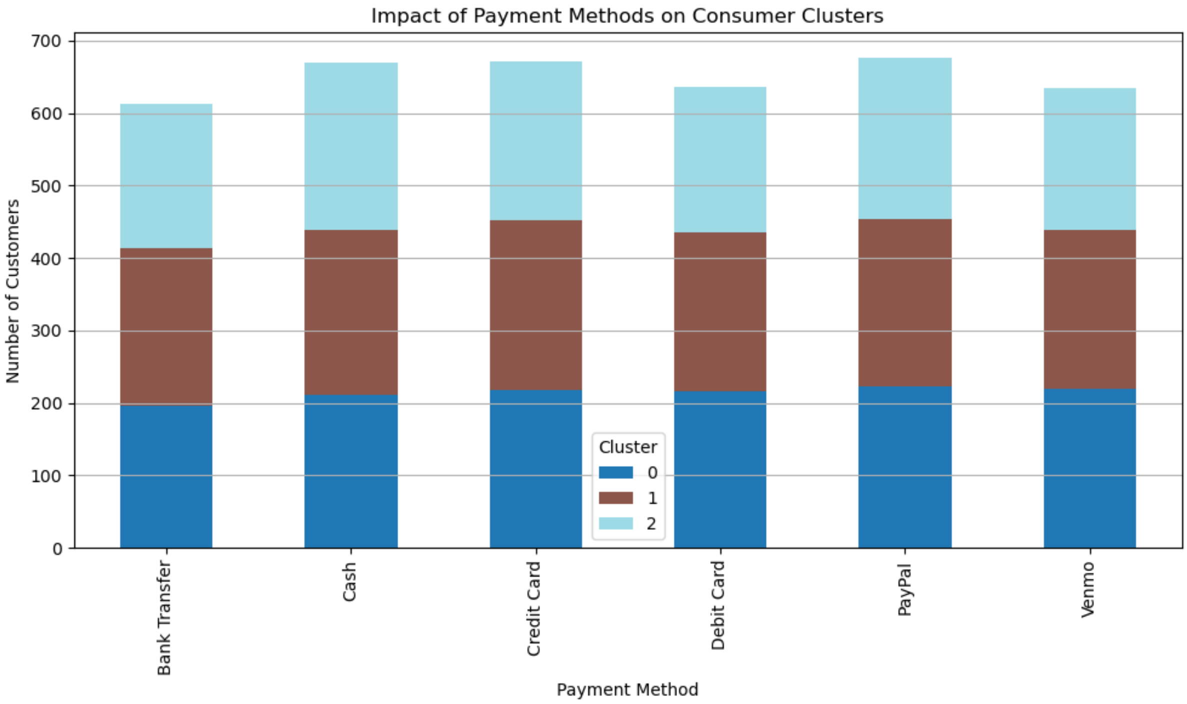


Figure 3.2.1.1 Bar Chart of K-Means clustering

The stacked bar chart, which is divided into three clusters, shows the relationship between various payment methods and consumer purchasing behaviour. Low-frequency, low-income customers are represented by Cluster 0, mid-frequency, higher-income buyers by

Cluster 1, and high-frequency but lower-income buyers by Cluster 2. Although there are minor variations, the distribution is generally balanced across all payment types. Payment Method 4, for instance, has a little greater concentration of Cluster 1 clients, indicating that it is more well-liked by moderately wealthy consumers. In contrast, Payment Method 0 seems to draw a little more Cluster 0 users, suggesting that it would be in line with infrequent, frugal shoppers. Businesses might use this information to customise promotions or preferred payment methods according to the type of customer.

```
📊 Cluster Interpretation Table:
         Purchase Amount (USD)  Frequency of Purchases  Previous Purchases  \
Cluster
0                         0.30                    1.26               23.26
1                         0.30                    3.00               24.89
2                         0.31                    4.68               27.96

         Customer Income Level
Cluster
0                         0.48
1                         2.00
2                         0.51
```

Figure 3.2.1.1 Result trained using K-Means clustering

The Cluster Interpretation Table provides information into three different behaviour segments. With a low-income level of 0.48 and a low purchase frequency of just 1.26, Cluster 0 is made up of low-frequency, low-income buyers who may be more price-sensitive. With the greatest income level (2.00), a moderate frequency of purchases (3.00), and somewhat greater prior purchases, Cluster 1 is clearly the higher-income group. This probably indicates more stable, financially capable consumers who make regular but restrained purchases. With the highest frequency of purchases (4.68) and the greatest number of prior purchases (27.96), Cluster 2 exhibits the most active shopping behaviour; yet, their income level is comparatively low (0.51). This suggests a group of regular, devoted customers who may be frugal but always involved. Based on their behaviour and financial situation, these results assist in segmenting customers for more focused promotions.

### 3.2.2 DBSCAN

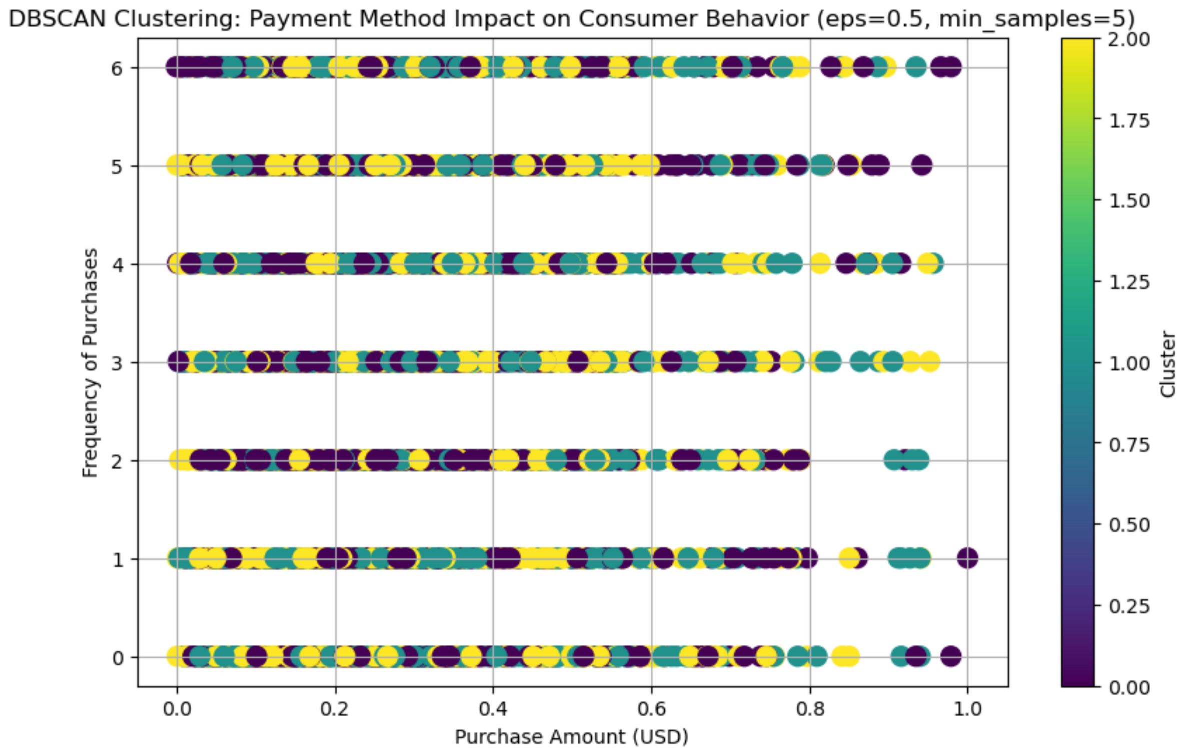


Figure 3.2.2.1 Scatter plot of DBSCAN clustering

The DBSCAN clustering visualization highlights how consumer behaviour, based on purchase amount and frequency, forms distinct groupings when analysed with parameters eps=0.5 and min_samples=5. Three major clusters (designated 0, 1, and 2) are visible in the plot, suggesting that customers frequently follow particular buying habits. For example, whilst some clusters comprise more diverse or sparse behaviours, others indicate high-frequency, low-amount purchases. The existence of several dense groupings raises the possibility that payment methods may have an impact on particular behavioural segments, maybe revealing consumer preferences or limitations. The dataset forms well-defined groupings under the current DBSCAN parameters, as evidenced by the successful cluster assignment of all data points and the absence of noise.

```
Number of points in each DBSCAN cluster (including noise):
DBSCAN Cluster
0    1349
2    1295
1    1256
Name: count, dtype: int64

Number of noise points: 0

Average values per cluster:
                Purchase Amount (USD)  Frequency of Purchases  \
DBSCAN Cluster
0                                0.30                    3.00
1                                0.31                    2.98
2                                0.30                    2.94

                Previous Purchases  Customer Income Level
DBSCAN Cluster
0                            24.89                    2.0
1                            25.66                    1.0
2                            25.53                    0.0

DBSCAN Cluster distribution by Payment Method:
DBSCAN Cluster        0    1    2
Payment Method Name
Bank Transfer       218  195  199
Cash                227  214  229
Credit Card         234  211  226
Debit Card          219  210  207
PayPal              231  223  223
Venmo               220  203  211
```

Figure 3.3.2.2 Result trained using DBSCAN clustering

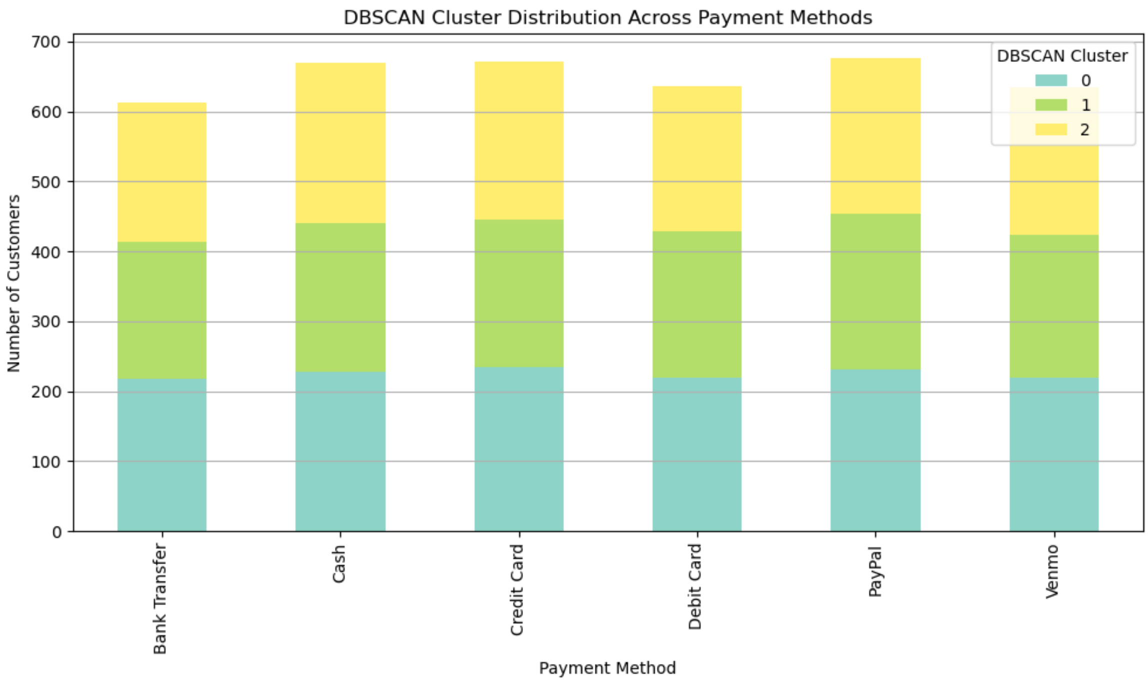


Figure 3.3.2.3 Bar Chart of DBSCAN clustering

With no noise points found, the DBSCAN clustering analysis separated the consumer data into three separate clusters (0, 1, and 2). This suggests that the data is well-structured for clustering. With 1,349 consumers in Cluster 0, 1,295 in Cluster 2, and 1,256 in Cluster 1, each cluster has a nicely balanced number of points.

Purchase behaviour appears to be consistent among clusters, as seen by the fact that, on average, all clusters exhibit extremely comparable purchasing behaviour in terms of purchase frequency (about three times) and purchase value (around \$0.30 to \$0.31). They vary considerably, though, in terms of prior purchases and client income levels:

- Cluster 0 has a middle-income level (2) and a moderate history of prior purchases (24.89).
- Cluster 1, which is associated with income level 1, had the most prior purchases (25.66).
- Cluster 2 is associated with the lowest income level (0) and has a significantly lower average of prior purchases (25.53).

The use of various payment methods is very evenly spread across all clusters when split down by cluster; no technique exhibits a clear preference for one over another. This suggests that other behavioural or demographic characteristics, such as income level and previous shopping patterns, have a stronger influence on the clustering than payment method. All things considered, DBSCAN shows modest but significant segmentation in patterns of customer behaviour.

## 3.3 Impact of Customer Income Level on Purchasing Behaviour

Customer income level is crucial in determining purchasing behaviour. By analysing how income level relates to purchasing frequency, along with other factors like purchase amount, age, and previous purchases, we can gain a deeper understanding of consumer decision-making. This clustering analysis focuses on the relationship between income and purchasing frequency and some additional features to better understand customer purchasing patterns.

### 3.3.1 K-Means Clustering


Figure 3.3.1.1 Scatter plot of K-means clustering

From the scatter plot, we can observe the distribution of customer clusters based on Purchase Amount (USD) and Frequency of Purchases. The plot clearly illustrates how the clusters differ in terms of spending and purchase frequency. Cluster 0, represented by the purple colour, contains customers with low spending and infrequent purchases. Cluster 1, marked by green, groups customers with moderate spending and a moderate frequency of purchases. Finally, Cluster 2, in yellow, represents customers with relatively higher spending and frequent purchases. The plot visually captures the distinct purchasing behaviours of each cluster, highlighting how customers' income levels influence their spending patterns.

```
Cluster Centers (Scaled Values):
[[0.24007782 0.30445246 0.21102464 0.47604011 0.45665052]
 [1.         0.29891551 0.50049419 0.50466157 0.48760231]
 [0.25236967 0.30417337 0.77988415 0.52340807 0.54787697]]

Dataset with Cluster Labels:
   Customer ID  Customer Income Level  Purchase Amount (USD)  \
0            1                      2               0.250559
1            2                      1               0.431133
2            3                      0               0.118262
3            4                      2               0.127593
4            5                      2               0.365736

   Frequency of Purchases  Age  Previous Purchases  Cluster
0                       3   55                  14        1
1                       3   19                   2        0
2                       6   50                  23        2
3                       6   21                  49        1
4                       0   45                  31        1
```

Figure 3.3.1.2 Result of K-means clustering

From the result, the cluster centres show the average value for each feature in every group. Cluster 0 has an average Customer Income Level of 0.2401 (indicating low-income), average Purchase Amount (USD) of 0.3045, and average Frequency of Purchases of 0.2110, which show customers who spend moderately but purchase infrequently. For Cluster 1, with average Customer Income Level of 1.0000 (indicating high-income), average Purchase Amount (USD) of 0.2989, and average Frequency of Purchases of 0.5005, represents high-income customers who make moderate purchases at a moderate frequency. Cluster 2 has an average Customer Income Level of 0.2524 (moderate-income), average Purchase Amount (USD) of 0.3042, and average Frequency of Purchases of 0.7799, which indicate the customers who make frequent purchases with moderate spending. The cluster labels show how customers are grouped according to their income level, frequency of purchases, and spending patterns. For example, a customer with income level 2 (high income), purchase amount 0.25 (moderate spending), frequency of purchase 3 (moderate purchase frequency) will be clustered into cluster 1, which represents high-income customers with moderate spending and moderate purchase frequency.

### 3.3.2 DBSCAN


Figure 3.3.2.1 Scatter plot of DBSCAN clustering

From the scatter plot, we can observe the clusters based on Purchase Amount (USD) and Frequency of Purchases. The plot clearly shows how the clusters differ in terms of spending and purchase frequency. Cluster 0, represented by the purple colour, contains customers with low spending and infrequent purchases. Cluster 1, marked by green, groups customers with moderate spending and a moderate frequency of purchases. Finally, Cluster 2, in yellow, represents customers with relatively higher spending and frequent purchases. The plot visually captures the distinct purchasing behaviours of each cluster, highlighting how customers' income levels influence their spending patterns. The clustering was performed using DBSCAN with eps=0.5 and min_samples=5, where eps define the maximum distance between two points for them to be considered neighbours, and min_samples defines the minimum number of points required to form a dense region (cluster). These parameter values allowed DBSCAN to effectively identify clusters and exclude noise.

```
Number of points in each DBSCAN cluster (including noise points):
DBSCAN Cluster
0    1349
2    1295
1    1256
Name: count, dtype: int64

Number of noise points: 0

Average values for each cluster (for insight into cluster centers):
                Customer Income Level  Purchase Amount (USD)  \
DBSCAN Cluster
0                                 2.0               0.298916
1                                 1.0               0.307106
2                                 0.0               0.301606

                Frequency of Purchases        Age  Previous Purchases
DBSCAN Cluster
0                             3.002965  44.242402           24.892513
1                             2.981688  43.343153           25.661624
2                             2.938996  44.590734           25.528958

Dataset with DBSCAN Cluster Labels:
   Customer ID  Customer Income Level  Purchase Amount (USD)  \
0            1                      2               0.250559
1            2                      1               0.431133
2            3                      0               0.118262
3            4                      2               0.127593
4            5                      2               0.365736

   Frequency of Purchases  Age  Previous Purchases  DBSCAN Cluster
0                       3   55                  14               0
1                       3   19                   2               1
2                       6   50                  23               2
3                       6   21                  49               0
4                       0   45                  31               0
```

Figure 3.3.2.2 Result of DBSCAN clustering

From the result, we can see the number of points in each DBSCAN cluster, where Cluster 0 has 1349 points, Cluster 1 has 1295 points, and Cluster 2 has 1256 points. There are 0 noise points, meaning all data points have been assigned to a cluster. The average values for each cluster show that Cluster 0 is characterized by high-income customers (average income level = 2.0) who have moderate spending (average purchase amount = 0.2989) and infrequent purchasing behaviour (average frequency of purchases = 3.0). Cluster 1 represents medium-income customers (average income level = 1.0) with moderate spending (average purchase amount = 0.3071) and moderate purchase frequency (average frequency of purchases = 2.98), while Cluster 2 consists of low-income customers (average income level = 0.0) who make frequent purchases (average frequency of purchases = 2.94) with moderate spending (average purchase amount = 0.3016). The dataset with the DBSCAN cluster labels shows how each customer is grouped based on their spending habits, providing useful insights into the relationship between income levels and purchasing behaviour. For example, a customer with income level 1 (medium income), purchase amount 0.43 (moderate spending), and frequency of purchase 3 (moderate purchase frequency) will be clustered into Cluster 0, which represents high-income customers with moderate spending and moderate purchase frequency.

# 4.0 Model Evaluation

| Criteria | Description |
|---|---|
| **Intra-cluster similarity** | How close members of the same cluster are |
| **Inter-cluster separation** | How far apart the clusters are from each other |
| **Compactness** | Small average distance from members to centroid |
| **Noise handling / Outlier detection** | Ability to ignore irrelevant data |
| **Shape detection** | Ability to detect arbitrary shapes |
| **Interpretability** | Ease of understanding and using clusters for business |
| **Davies-Bouldin Index (DBI)** | Lower DBI = better clustering (only for K-Means) |

4.0.1 Description Table

This section presents the evaluation and comparison of the two clustering models used in this analysis: K-Means and DBSCAN. Clustering quality should be assessed using internal validation techniques, such as intra-cluster similarity, inter-cluster separation, and compactness. When available, metrics like the Davies-Bouldin Index (DBI) further aid in assessing how well the clusters are separated and internally cohesive.

## 4.1 Age Group Dominance

| Evaluation Metric | K-Means | DBSCAN |
|---|---|---|
| Cluster Count | 3 clusters | 7 clusters |
| Cluster Shape | Spherical | Arbitrary |
| Intra-cluster similarity | Good | Very high |
| Inter-cluster separation | Clear separation | lower inter-cluster separation |
| Outlier Handling | No | Effectively handles outliers |
| Interpretability | High – Younger, middle, older | Moderate interpretability |

K-Means is the better model for age group segmentation due to its effective separation, low DBI, and business-usable groupings. DBSCAN, with 7 clusters, offers a simpler density - based grouping, useful if only broad age groupings are needed, but lacks the detailed breakdown of K-Means.

## 4.2 Payment Method

| Evaluation Metric | K-Means | DBSCAN |
|---|---|---|
| Cluster Count | 3 | 3 |
| Cluster Shape | Spherical | Arbitrary |
| Intra-cluster similarity | Moderate to good | Moderate |
| Inter-cluster separation | Reasonable | Moderate |
| Outlier Handling | No | Yes |
| Interpretability | High | Moderate |

In analysing payment method preferences, both K-Means and DBSCAN successfully formed three customer segments. However, K-Means produced more interpretable clusters aligned with business traits such as income level and purchase frequency. K-Means effectively identified low-income, low-frequency buyers; high-income, moderate buyers; and frequent but budget-conscious shoppers, making it ideal for targeted marketing strategies. DBSCAN also revealed similar patterns and handled data density well, but the behavioural distinctions among its clusters were less pronounced, and no noise points were detected. Overall, K-Means is the preferred model due to its clearer segmentation and better alignment with strategic decision-making.

## 4.3 Customer Income Level

| Evaluation Metric | K-Means | DBSCAN |
|---|---|---|
| Cluster Count | 3 | 3 |
| DBI | Not reported | Not applicable |
| Cluster Shape | Spherical | Arbitrary |
| Intra-cluster similarity | Moderate | Moderate |
| Inter-cluster separation | Moderate | Moderate |
| Outlier Handling | No | Yes |

In evaluating the impact of customer income level on purchasing behaviour, both K-Means and DBSCAN proved to be valid and effective clustering models. K-Means offered clear and interpretable cluster centroids, allowing straightforward identification of customer segments based on income, purchase amount, and purchase frequency, making it highly suitable for actionable business decisions. DBSCAN, while not centroid-based, successfully grouped customers by density and was particularly effective in capturing natural clusters that may not follow linear boundaries. Although both models performed well, K-Means holds a slight advantage due to its simplicity, ease of interpretation, and practical applicability in business contexts.

## 5.0 Conclusion

Based on the results get from the clustering models, K-Means and DBSCAN that applied to the customer shopping behaviour on fashion dataset, it can be concluded that both algorithms provide valuable insights to customer segmentation, even though with distinct strengths. The K-Means clustering provides well-separated and interpretable clusters, particularly useful for segmenting customers based on demographic characteristics like age and income. Its clear and actionable insights like identifying low-income, low-frequency buyers or high-income, moderate-frequency buyers, make it particularly beneficial for targeted marketing strategies. On the other hand, DBSCAN great at handling noisy data and identifying clusters based on density, but its results tend to be less interpretable and may not align as well with business strategies. Overall, K-Means is the better clustering method for this dataset due to its simplicity, clear segmentation, and stronger alignment with actionable business insights, while DBSCAN remains a valuable tool for handling outliers and uncovering less defined clusters.

## 6.0 References


1. Ibrahim, R. (2023). Data Preparation using RapidMiner [Lecture notes]. SECPH 2753: Data Mining. Universiti Teknologi Malaysia, Faculty of Computing.

2. **IBM Cloud Education** (2020). *K-Nearest Neighbors (KNN) Algorithm for Machine Learning*. IBM. https://www.ibm.com/cloud/learn/k-nearest-neighbors

3. **Scikit-learn Documentation** (2023). *Naïve Bayes*. scikit-learn. https://scikit-learn.org/stable/modules/naive_bayes.html

4. **Google Developers** (2021). *Decision Trees: How to Optimize for Business Decisions*. Machine Learning Crash Course. https://developers.google.com/machine-learning/decision-forests/decision-trees

# 7.0 Reflection

## 7.1 LEE YIN SHEN

### I. Project Reflection

In Project 1, we used classification methods like K-NN, Naïve Bayes, and Decision Trees to analyse the customer data in fashion industry. We aim to predict the factors like age group, payment method preference, and income level. To ensure the models work effectively, we first cleaned the dataset, removed unnecessary columns, transformed categorical data into numerical values, and split the data into training and testing sets. Then, the final dataset was used for the classification algorithm we chose to predict the related result.

In Project 2, we applied clustering methods like K-Means and DBSCAN to group customers based on their purchasing behaviour. For this part, we focus on clustering the customer data into meaningful clusters based on features like age, payment method, and income level. The clustering methods allowed us to explore natural groupings in the data without predefined labels that give us insights into customer behaviours.

In Project 1, I'm the person in charge for the data preparation process, and I was very confused for this part because sometimes, I don't know what type of process should I apply to some column. But finally, after the discussion with my teammate, I complete it, although we are not sure is the process correct. Overall, I learned the importance of data preparation for getting accurate results in this part.

In Project 2, I found that RapidMiner was difficult to use for clustering, because the result I got from it was a bit weird and not logic, even though I already spent hours in adjusting the connection between each component inside the tool. Then, I change to use python for this part for both data preprocessing and clustering, and things became much easier, and we were able to proceed smoothly.

The things that I learned from both projects was learning how to prepare a dataset properly, and also gained experience in using classification and clustering methods to analyse data. Moreover, I improved my skills in Python for data analysis, which will be useful for future projects.

## II. Course Reflection

Overall, I feel that I'm not fully met my personal learning expectation from this course. Before the course, I hoped that by the end of the course, I would able to manage a large dataset, clean them effectively, and create nice visualizations for it. However, the course mainly focused on the theory and methods of data mining, and actually did not provide hands-on experience on coding or using data mining tools to process a dataset. As a result, I still need to learn and practice these skills on my own.

From the course, I gained valuable knowledge especially from the project. I learned how to prepare a dataset from raw data, including cleaning and transforming it for analysis. I also gained experience using RapidMiner and Python to apply classification and clustering algorithms, which I can build upon with more practice.

As a data engineering student, working with data is a key part of my future career. So, I plan to apply the skills learned in this course to handle and process data more efficiently in the future projects. Overall, I will continue to practice and improve my data mining skills to better manage and derive insights from data in both academic and future career.

## 7.2 ELIJAH SHE YU SHENG

### I. Project Reflection

In Project 1, I was responsible for building the classification models such as K-NN, Naïve Bayes, and Decision Trees to analyse Age Group Dominance. As it was my first time working on a data mining project, I initially felt confused because I wasn't always sure if I was applying the methods correctly. However, after discussing with Yin Shen and Li Xin, I managed to find the right approach to solve the problem, although it took me some time to complete it. Overall, I learned how to use RapidMiner to generate classification results and also understood how it compares to running similar analyses using Python.

In Project 2, we realised that using RapidMiner for clustering was challenging. When I applied K-Means clustering, it worked, but using DBSCAN produced unexpected results. After discussing with my teammates, we decided to switch approaches. We had spent several hours trying to solve it in RapidMiner, but eventually, Yin Shen suggested using Python for both data preprocessing and clustering, which made the process much smoother. Through this, I also started learning how to use Python as another platform to analyse datasets efficiently.

From both projects, I learned how to apply classification and clustering techniques to analyse data using both RapidMiner and Python. Additionally, I improved my Python programming skills through these practical experiences.

### II. Course Reflection

Overall, I feel that I need to study much more in this course, as I realise that data mining is just a basic starting point for analysing datasets. I still have a lot to learn during lectures so that I can apply different machine learning techniques to analyse data effectively and obtain accurate results. Throughout this course, I learned more about using software applications to analyse data, which is very different from manually recording and processing data by hand to generate the results I need. Therefore, I believe that by continuing to learn and practise, I will be able to provide better analyses in the future.

## 7.3 NEO LI XIN

### I. Project Reflection

During Project 1, the part we struggled the most is deciding which and how to use the selected application to interpret our dataset. But thanks to Lee Yin Shen's and Elijah's immense effort, they made it easier for me to understand what to do with the application so that I can apply K-NN, Naive Bayes and Decision Tree models easily for the business insight evaluation (Most Used Payment Methods in Fashion Industry). From understanding what each operator does, how to connect each operator and the required settings for each operator, it's a tough process but still we managed to produce results with different models so that we can do a complete comparison.

During Project 2, we planned to use RapidMiner again from Project 1. However, unlike classification, we realized that doing clustering prove to be harder and challenging. So, after Lee Yin Shen suggestion, we switched to Python as it gives more simpler and direct results as well as more clean data visualization. Things went smoothly since, and we were able to complete our Project 2 with the help of Python. This experience taught me that there's always a better tool for different project requirements.

From both these projects, I discovered the use of different applications in performing data classification and data clustering. I don't think that I mastered the use of RapidMiner or Python yet, but I have built a foundation of how these tools can be applied for my future data related works.

### II. Course Reflection

During this course I learned a lot of different data mining techniques, including: data cleaning, classification models, clustering, and association. The particular interest are the models we used during the project: K-NN, Naive Bayes and Decision Tree Model. The course mostly focused on theory, however I wished this course covers more about the application and we can use it on the database. Overall, I really appreciate that this course teaches majority of the data mining process and the different models available to us.